

Use-Case Flow Specification

Hyperlocal Courier Dispatch & Tracking Engine

Use Case: Dispatch and Track Courier Delivery

Actors: Sender Client, Delivery Rider

Preconditions

- The sender can submit a courier request with pickup and destination details.
- Delivery riders can provide their active status and location.
- The courier request is valid and has not been completed or cancelled.

Postconditions

Success: A suitable rider is assigned, the parcel is picked up and delivered, live tracking is provided to the sender, the recipient's OTP is verified, and the delivery is marked as completed.

Failure: If no suitable active rider is available within the required radius, the request remains unassigned and the sender is informed.

Main Success Scenario

1. The Sender Client creates a courier request with pickup and destination details.
2. The system validates the request and finds the nearest active rider within a 3 km radius.
3. The system sends the pickup location and dispatch notification to the selected Delivery Rider.
4. The Delivery Rider accepts the request and the system assigns the delivery.
5. If multiple stops exist, the system generates an optimized delivery route.
6. The rider picks up the parcel and starts the delivery.
7. The system continuously sends the rider's GPS location and delivery status to the Sender Client.
8. The rider reaches the destination and requests the delivery OTP.
9. The recipient provides the OTP and the system verifies it.
10. If the OTP is valid, the system marks the parcel as delivered and notifies the Sender Client.

Alternate Flow: No Active Rider Within 3 km

1. The system searches for active riders within a 3 km radius.
2. No eligible rider is found.
3. The system does not assign an offline or out-of-range rider.
4. The request remains pending or unassigned.
5. The Sender Client is informed that a suitable rider is currently unavailable.

Summary

This use case covers courier request creation, nearby rider assignment, route optimization, live tracking, and OTP-based delivery verification from request initiation through successful delivery.